Soumadeep Saha, PhD

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Summary

My current research at *ANITI* explores techniques to better understand *reasoning with large language models*. My PhD research centered on enhancing deep learning systems with the strengths of symbolic AI techniques for greater logical coherence and domain constraint adherence. While my primary focus is *natural language processing*, I actively collaborate across domains including vision, medicine, biology, astrophysics, and business applications.

Education

Indian Statistical Institute, Kolkata — Kolkata, India
PhD (Computer Science) — *2020–2025*
- Supervisor: **Prof. Utpal Garain**
- Dissertation: **"Domain Obedient Deep Learning"**.

Indian Institute of Science Education and Research, Kolkata — Kolkata, India
Integrated BS & MS in Physics (Minor in Mathematics) — *2015–2020*
- Master's dissertation: **"Towards Robust Deep Learning Systems"**.

Selected Works

- BODHI: Do LLMs Branch Out and Discover Heterogeneous Inferences? — Preprint
  S. Saha, K. Sharma, A. Chaturvedi, N. Asher — 10.48550/arXiv.2608.02867
- TAPIOCA: Why Task-Aware Pruning Improves OOD model Capability — Preprint
  K. Sharma, O. Naim, **S. Saha**, N. Asher — 10.48550/arXiv.2605.14738
- The Synergistic Integration of Access Control Management and Large Language Model Agents: A Survey — Preprint
  F. Periti, **S. Saha** — 10.36227/techrxiv.177075184.43798334/v1
- KisMATH: Do LLMs Have Knowledge of Implicit Structures in Mathematical Reasoning? — **TACL** (2026), 14: 1308-1328
  S. Saha, A. Chaturvedi, S. Saha, U. Garain, N. Asher
- sudoLLM: On Multi-role Alignment of Language Models — **EMNLP 2025** (findings)
  S. Saha, A. Chaturvedi, J. Mahapatra, U. Garain
- Language Models are Crossword Solvers — **NAACL 2025** (main)
  S. Saha, S. Chakraborty, S. Saha, U. Garain
- Analyzing Semantic Faithfulness of Language Models via Input Intervention on Conversational Question Answering — **Computational Linguistics** (2024), 50(1): 119-155
  A. Chaturvedi, S. Bhar, **S. Saha**, U. Garain, N. Asher
- On Measuring Intrinsic Causal Attributions in Deep Neural Networks — **CLeaR 2025**
  S. Saha, D. V. Rathore, **S. Saha**, D. Doermann, U. Garain.
- VALUED - Vision and Logical Understanding Evaluation Dataset — **DMLR** (2024), (13):1-18.
  S. Saha, S. Saha, U. Garain
- MedTric : A clinically applicable metric for evaluation of multi-label computational diagnostic systems — **PLOS One** (2023), 18(8): e0283895
  S. Saha, U. Garain, A. Ukil, A. Pal, S. Khandelwal.

Patents

- Method and System for Contradiction Avoided Learning for Multi-Class Multi-Label Classification — **US Patent - US12038949B2**
  S. Saha, U. Garain, A. Ukil, A.Pal — Granted Jul. 2024
- Method and System for Evaluating Clinical Efficiency of Multi-Label Multi-Class Computational Diagnostic Models — *Pending*
  U. Garain, **S. Saha**, A. Ukil, T. Deb, S. Richa A. Pal, S. Khandelwal — Filed Sep. 2022.

Grants, Fellowships & Awards

OpenAI Grant
Jan '25

Awarded $5000 in computational credits from **OpenAI** under their *Researcher Access Program* to support my work on LLM alignment.

Helmholtz Visiting Researcher Grant
Jul '24 – Sep '24

Awarded the Helmholtz Information and Data Science Academy (HIDA) *visiting researcher grant* to work at the Institute of Aerospace Medicine, DLR (German Aerospace Center). Conducted research on improving AI-based retinal diagnostics for spaceflight applications.

KVPY & NTSE
2015, 2013

Awarded the prestigious Kishore Vigyan Pratoshan Yojna (**KVPY**) fellowship from the Department of Science and Technology, Govt. of India, and recipient of the National Talent Search Examination (**NTSE**) Scholarship from NCERT, Govt. of India.

Experience

Alleima
Mar '24 – May '24

Developed a custom end-to-end computer vision-based solution for assembly-line object detection/tracking, enabling autonomous logging of crucial data with existing infrastructure.

TCS Research
Nov '21 – Jul '22

Conducted research as part of a team on ECG-based diagnosis of cardiovascular diseases. Independently identified key gaps in the literature and developed innovative, state-of-the-art solutions to address them, resulting in two patents, and publications.

Talks and Presentations

- ❏ Organized a 4-day workshop for DataLab, **Capital One**, Bangalore (2023).
- ❏ Subject matter expert for Be10x; designed MOOCs, and delivered live seminars to 500+ participants (2024).
- ❏ TA for Natural Language Processing course at ISI Kolkata (2022–2025).
- ❏ Instructor at the Winter School of Deep Learning (WSDL), ISI Kolkata (2021–2024).
- ❏ Instructor for the Comprehensive Course on Business Analytics, ISI Kolkata (2022).
- ❏ Presented my work on Logically Coherent Deep Learning at Amazon Research Days (ARD '22).
- ❏ Presented my work on domain-obedient self-supervision at W3PHIAI, AAAI 2024, Vancouver, Canada.
- ❏ Presented my work on constrained LLM inference at NAACL 2025, Albuquerque, USA.

Technical Skills

❏ **Languages & Frameworks**: Python (`PyTorch`, `HuggingFace`), C++, bash.
❏ **Infrastructure & Compute**: Multi-node and multi-GPU distributed (`FSDP`, `DeepSpeed`) training (RLVR, SFT) on GPU clusters.
❏ **LLMs & Alignment**: RLVR, RLHF, DPO, PEFT/Q-LoRA, RAG, In-context Learning.
❏ **Computer Vision**: VLMs, Vision Transformers (ViTs), CNNs, Object Detection & Segmentation.

Interests

❏ **Robotics/DIY**–Active interest in 3D printing, homelabbing, robotics, electronics, etc. I have conducted introductory workshops on robotics and was the Secretary of the Robotics and Astronomy Club at IISER Kolkata.
❏ **Sports**–Represented my college in national-level sports meets in basketball and volleyball. Played in my state's Senior Division Men's Basketball League.
❏ **Music**–Classically trained pianist.
